



Quiz3

Team project: Tic-tac-toe

Tic-tac-toe is a common board game played on three-in-a-row boards and it can be traced back to ancient Egypt where such game boards have been found on roofing tiles dating from around 1300 BC. You can learn more about the game from this link, below.

<https://en.wikipedia.org/wiki/Tic-tac-toe>

Task:

Create an interactive Tic-tac-toe game using java.

Implementation:

- The game has two modes. Mode 1 is a game between two players. Mode 2 is one player against the CPU.
 - Once the program runs, the following message is displayed
 - Enter 1 for 2 players, 2 for playing against the CPU or 0 to quit.
 - Once a mode is selected, a plain board is displayed in the console with tiles numbered 1 thru 9 (see the example below)
 - The board is drawn using three characters only -, + and | (that is a dash, a plus sign and a pipe) as seen below in the example.
 - The size (width and height) of the board is up to you.
 - The code should display whose turn it is by displaying the message; “player 1 turn: “ when it’s player 1 turn and “player 2 turn:” message when it’s player 2 turn.
 - The player(s) should only enter the number of the tile they want
 - Once the player input the “valid” tile number, the code redisplay the board with the proper tiles filled with Xs for player 1 and Os for player 2 (see the example below)
 - In case of mode 2, once the player input the “valid” tile number, the code generate CPU move “trying to win the game” then redisplay the board with the proper tiles filled with Xs for player 1 and Os for CPU then it is the player’s turn to make the next move.

Winning messages

- If player 1 reaches three-in-a-row first, your code should print the message
 - Game Over! Player 1 Wins. then your logic should exit the game
- If player 2 reaches three-in-a-row first, your code should print the message
 - Game Over! Player 2 Wins, then your logic should exit the game
- If CPU reaches three-in-a-row first, your code should print the message
 - Game Over! CPU Wins, then your logic should exit the game
- If there no valid moves, and the game tend to tie in either mode, your code should print the message: “Game Over! It is tie”



Error validation

- Your game should allow each player to enter only valid number between 1 and 9 that represents an available (not occupied) tile.
 - Any incorrect value entered by the player, your code should print a message "incorrect entry, please try again"
 - A player has three consecutive incorrect entries and total 5 incorrect nonconsecutive entries to be considered forfeiting the game. In his case, the code will print the message "Player1(or2) forfeit the game due to reaching maximum incorrect entries."
 - If a player enters 0 at any stage of the game, the player forfeits the game to the other player or to the CPU.

Submittal

- Your project must have a class named TicTacToe.java with a main method.
 - You are allowed to create any number of methods you like to help organize your code.
 - You are allowed to create any additional classes as long as they don't have main methods.
 - All classes must be zipped in a single zip file.
 - Upload one zip file per each group. The file name should be Quiz3.student1LastName.student2LastName.zip

Examples:

Mode1	Mode2
Enter 1 for 2 players mode or enter 2 for playing against the CPU or enter 0 to quit. 1 <your code displays> 1 2 3 -----+-----+----- 4 5 6 -----+-----+----- 7 8 9 Player 1 turn: 5 <your code displays>	Enter 1 for 2 players mode or enter 2 for playing against the CPU or enter 0 to quit. 2 <your code displays> 1 2 3 -----+-----+----- 4 5 6 -----+-----+----- 7 8 9 Player 1 turn: 5 <your code displays player1 move and CPU move as well>

```

1 | 2 | 3
---+---+---
4 | x | 6
---+---+---
7 | 8 | 9

```

Player 2 turn: 9
<your code displays>

```

1 | 2 | 3
---+---+---
4 | x | 6
---+---+---
7 | 8 | 0

```

Player 1 turn: 2
<your code displays>

```

1 | x | 3
---+---+---
4 | x | 6
---+---+---
7 | 8 | 0

```

Player 2 turn: 6

```

1 | x | 3
---+---+---
4 | x | 0
---+---+---
7 | 8 | 0

```

Player 1 turn: 10
<your code displays>

Incorrect entry, please try again.

Player 1 turn: 8
<your code displays>

```

1 | 2 | 3
---+---+---
4 | x | 6
---+---+---
7 | 8 | 0

```

Player 1 turn: 2
<your code displays player1 move and CPU move as well>

```

1 | x | 3
---+---+---
4 | x | 0
---+---+---
7 | 8 | 0

```

Player 1 turn: 10
<your code displays>

Incorrect entry, please try again.

Player 1 turn: 8
<your code displays player1 move and CPU move as well>

```

1 | x | 3
---+---+---
4 | x | 0
---+---+---
7 | x | 0

```

Player 1 Wins!



<pre>1 x 3 ---+---+--- 4 x O ---+---+--- 7 x O</pre> Player 1 Wins!	
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Grading

Code doesn't compile	0
CPU doesn't win in a clear win move	-10
Either of the players get three in a row and the logic doesn't declare a winner	-10
Game ends in a tie but the logic doesn't declare it as a tie	-10
Incorrect runnable file name	-5
Incorrect zip file name	-5
Incorrect error message(s)	-5
No comments in the code	-5
Distorted board display (lines are not horizontally or vertically aligned)	-10
Incorrect board representation of player's move	-10

Good Luck!

Dr. Nassar